Yue Jiang

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Website: https://yuejiang-nj.github.io/
LinkedIn: www.linkedin.com/in/yuejianguoft
Google Scholar: shorturl.at/zDN56

By developing models to better understand the structure and semantics of digital interfaces alongside models to simulate how people perceive, navigate, and process visual information, my research vision is to develop human-centered technologies that enhance human capabilities by unifying computational interface understanding with predictive models of human attention and behavior. This integration allows us to build adaptive, accessible, and intelligent systems that dynamically reshape themselves around the user's intent, abilities, and context.

Current Position

Nov 2025 - **Kahlert School of Computing, University of Utah, USA**
Assistant Professor
- **NVIDIA Academic Grant Program Award**
- **Institute Council Member of the Institute for Digital Experience**
- **Committee Member of Undergraduate AI Major Establishment Committee**

Education

Oct 2025 **Aalto University & Finnish Center for Artificial Intelligence (FCAI), Finland**
Ph.D. in Electrical Engineering (Interactive Systems)
- **Aalto Best Dissertation Award**
- **Meta Research PhD Fellowship** (21 worldwide selected from 3,200+ applicants)
- **Google Europe Students with Disabilities Scholarship** (10 in Europe)
- **Nokia Scholarship**
- **Heidelberg Laureate Forum Young Researcher**
- **UCSD Data Science Rising Star** (30 worldwide)
- **CHI2024 Doctoral Consortium**

Supervisors: Prof. Antti Oulasvirta and Prof. Vikas Garg

Jan 2024 - May 2024 **Human Computer Interaction Institute (HCII), Carnegie Mellon University, USA**
Visiting Ph.D. Student
Supervisor: Prof. Jeffrey Bigham

2020 **University of Maryland, College Park, USA**
Master of Science in Computer Science (Computer Graphics)
Supervisor: Prof. Matthias Zwicker

2017 **University of Toronto, Canada**
Honors Bachelor of Science in Computer Science Specialist and Mathematics Major (High Distinction) [Degree granted in 2018]
Supervisor: Prof. Gerald Penn

Faculty Grants

June 2026 – Apr 2027 **Responsible AI Seed Grants**
Title: Responsible AI-Driven Model Development for Proactive Household Environmental Exposure Assessment Using Low-Cost PM2.5 Sensors

PIs: Darrah Sleeth, Yue Jiang, Scott Collingwood, Nancy Daher, Katherine Sward

Total: $40,000

Mar 2026 – June 2026 **Basic Engineering Equipment Funds (BEEF)**

Title: Augmenting Student Learning with Immersive Interactive Experiences

PIs: Yue Jiang (**Lead PI**), Andrew McNutt, Sameer Patil, Jason Wiese, Eliane Wiese, Kate Isaacs, Marina Kogan

Total: $99,925

Mar 2026 **NVIDIA Academic Grant Program Award**

PI: Yue Jiang

Total: GPU Support

Professional Experience and Internships

June 2023 – Sept 2023 **Apple Inc.**, Seattle, Washington, USA

AI/ML Group Research Intern

Supervisor: Dr. Jeffery Nichols

- Developed a UI-centric instruction-following vision-language agent designed for performing UI-related tasks (Published at IUI2025).

Oct 2020 – Apr 2022 **Max Planck Institute for Informatics (MPII)**, Germany

Researcher

Supervisor: Prof. Christian Theobalt

- Proposed a human performance capture approach, which simultaneously captures human pose, clothing deformation, facial expression, and hand gestures from videos (Published at BMVC2022).

June 2020 – Aug 2020 **Carnegie Mellon University**, Pennsylvania, USA

Research Intern

Supervisor: Prof. Chris Harrison

- Developed a privacy-preserving activity recognition system based on Doppler radar data (Published at CHI2021).

Mar 2020 – Aug 2020 **Adobe Research**, College Park, Maryland, USA

Research Intern

Supervisor: Dr. Vlad Morariu

- Proposed a method for generating and optimizing dynamic content and layout to interactively adapt a document to various devices, author preferences, and viewer preferences. (Published at VL/HCC2024)

June 2019 – Aug 2019 **Shenzhen University**, Shenzhen, China

Visiting Graduate Research Student

Supervisor: Prof. Hui Huang & Prof. Daniel Cohen-Or

- Worked on differentiable rendering and sketching methods.

May 2017 – Apr 2018 **Intel Corporation**, San Jose, California, USA

Software Engineer

- Developed software tools for FPGA groups.

May 2016 – Apr 2017 **University of Toronto**, Toronto, Canada

Research Intern

Supervisor: Prof. Gerald Penn

- Worked on Lambek Categorial Grammar for practical parsing.

Full Paper Publications

Note: CHI, UIST, and CSCW are recognized as top-tier HCI conferences; CVPR and ECCV are top-tier Computer Vision conferences. VL/HCC is a prominent conference on programming tools, IUI is a top conference on AI for UI-related research, while ETRA is the largest conference focused specifically on eye tracking. The average acceptance rate for all of these conferences is approximately 25%.

[23] Eunhye Kim, DaEun Choi, Bryan Min, Hyunjung Yi, **Yue Jiang**, Juho Kim. 2026. Maru: Information Architecture as a Shared Language for Generating Aligned and Persistent User Interfaces. *In Proceedings of the ACM Symposium on User Interface Software and Technology (**UIST2026**)*. [PDF]

[22] Boyu Li, Linjie Qiu, Lin-Ping Yuan, Duotun Wang, **Yue Jiang**, Zeyu Wang, Hongbo Fu. Spatula: Exploring On-Demand In-Situ Interfaces and Interaction for Attribute Control. *In Proceedings of the ACM Symposium on User Interface Software and Technology (**UIST2026**)*. [PDF]

[21] Duotun Wang, Linjie Qiu, Linlin Cai, **Yue Jiang**, Mingming Fan, and Zeyu Wang. Visual Generative Artificial Intelligence (GenAI) in Mixed Reality (MR): A Survey of Techniques, Interaction Designs, and Evaluations. *International Journal of Human-Computer Interaction (**IJHCI2026**)*. [PDF]

[20] Zixin Guo*, **Yue Jiang* (co-first author)**, Luis Leiva, Antti Oulasvirta. Scanpath Prediction for Visual Search on Graphical User Interfaces. *In Proceedings of the 44th Annual SIGCHI Conference on Human Factors in Computing Systems (**CHI2026**)*. [PDF]

[19] **Yue Jiang**, Eldon Schoop, Amanda Swearngin, Jeffrey Nichols. ILuvUI: Instruction-tuned LangUage-Vision modeling of UIs from Machine Conversations. *The 30th Annual ACM Conference on Intelligent User Interfaces (**IUI2025**)*. [PDF]

[18] Aini Putkonen, **Yue Jiang**, Jingchun Zeng, Olli Tammilehto, Jussi P. P. Jokinen, Antti Oulasvirta. Understanding Visual Search in Graphical User Interfaces. *International Journal of Human-Computer Studies (**IJHCS2025**)*. [PDF]

[17] **Yue Jiang***, Zixin Guo*, Hamed Rezazadegan Tavakoli, Luis A. Leiva, Antti Oulasvirta. EyeFormer: Predicting Personalized Scanpaths with Transformer-Guided Reinforcement Learning. *In Proceedings of the ACM Symposium on User Interface Software and Technology (**UIST2024**)*. [PDF]

[16] Yi-Hao Peng, Faria Huq, **Yue Jiang**, Jason Wu, Amanda Li, Jeffrey P. Bigham, and Amy Pavel. DreamStruct: Understanding Slides and UIs via Programmable Synthetic Data Generation. *European Conference on Computer Vision (**ECCV2024**)*. [PDF]

[15] **Yue Jiang**, Changkong Zhou, Vikas Garg*, Antti Oulasvirta*. Graph4GUI: Graph Neural Networks for Representing Graphical User Interfaces. *In Proceedings of the 42nd Annual SIGCHI Conference on Human Factors in Computing Systems (**CHI2024**)*. [PDF]

[14] Maryam Taeb, Amanda Swearngin, Eldon Schoop, Ruijia Cheng, **Yue Jiang**, Jeffrey Nichols. AXNav: Replaying Accessibility Tests from Natural Language. *In Proceedings of the 42nd Annual SIGCHI Conference on Human Factors in Computing Systems (**CHI2024**)*. [PDF]

[13] **Yue Jiang**, Christof Lutteroth, Rajiv Jain, Christopher Tensmeyer, Varun Manjunatha, Wolfgang Stuerzlinger, Vlad Morariu. FlexDoc: Flexible Document Adaptation through Optimizing both Content and Layout. *In Proceedings of the IEEE Symposium on Visual Languages and Human-Centric Computing (**VL/HCC2024**)*. [PDF]

[12] Yao Wang, **Yue Jiang**, Zhiming Hu, Constantin Ruhdorfer, Mihai Bâce, Andreas Bulling. VisRecall++: Analysing and Predicting Visualisation Recallability from Gaze Bahaviour. *In Proceedings of the ACM on Human-Computer Interaction (**PACM-HCI**). Presented at the 2024 ACM Symposium on Eye Tracking Research and Applications (**ETRA2024**)*. [PDF]

[11] Parvin Emami, **Yue Jiang**, Zixin Guo, Luis A. Leiva. Impact of Design Decisions in Scanpath Modeling. *In Proceedings of the ACM on Human-Computer Interaction (**PACM-HCI**). Presented at the 2024 ACM Symposium on Eye Tracking Research and Applications (**ETRA2024**)*. [PDF]

[10] **Yue Jiang**, Luis A. Leiva, Hamed Rezazadegan Tavakoli, Paul R. B. Houssel, Julia Kylmala, Antti Oulasvirta. UEyes: Understanding Visual Saliency across User Interface Types. *In Proceedings of the 41st Annual SIGCHI Conference on Human Factors in Computing Systems (**CHI2023**)*. [PDF]

[9] **Yue Jiang**, Marc Habermann, Vladislav Golyanik, Christian Theobalt. HiFECap: High-Fidelity and Expressive Capture of Human Performances from Monocular Videos. *In Proceedings of the 2022 British Machine Vision Virtual Conference* (***BMVC2022***). [PDF]

[8] Jiao Sun, Tongshuang Wu, **Yue Jiang**, Ronil Awalegaonkar, Xi Victoria Lin, Diyi Yang. Pretty Princess vs. Successful Leader: Gender Roles in Greeting Card Messages. *In Proceedings of the 40th Annual SIGCHI Conference on Human Factors in Computing Systems* (***CHI2022 Best Paper Honorable Mention***). [PDF]

[7] **Yue Jiang**, Wolfgang Stuerzlinger, Christof Lutteroth. ReverseORC: Reverse Engineering of Resizable User Interface Layouts with OR-Constraints. *In Proceedings of the 39th Annual SIGCHI Conference on Human Factors in Computing Systems* (***CHI2021***). [PDF]

[6] Karan Ahuja, **Yue Jiang**, Mayank Goel, Chris Harrison. Vid2Doppler: Synthesizing Doppler Radar Data from Videos for Training Privacy-Preserving Activity Recognition. *In Proceedings of the 39th Annual SIGCHI Conference on Human Factors in Computing Systems* (***CHI2021***). [PDF]

[5] Zhicong Lu, **Yue Jiang**, Chenxinran Elise Shen, Margaret C Jack, Daniel Wigdor, Mor Naaman. Study of Perceptions of COVID-19 Misinformation in China. *In Proceedings of the 24th ACM Conference on Computer-Supported Cooperative Work and Social Computing* (***CSCW2021***). [PDF]

[4] **Yue Jiang**, Dantong Ji, Zhizhong Han, Matthias Zwicker. SDFDiff: Differentiable Rendering of Signed Distance Fields for 3D Shape Optimization. *Conference on Computer Vision and Pattern Recognition* (***CVPR2020 Oral (Top 5%)***). [PDF]

[3] **Yue Jiang**, Wolfgang Stuerzlinger, Matthias Zwicker, Christof Lutteroth. ORCSolver: An Efficient Solver for Adaptive GUI Layout with OR-Constraints. *In Proceedings of the 38th Annual SIGCHI Conference on Human Factors in Computing Systems* (***CHI2020***). [PDF]

[2] Zhicong Lu, **Yue Jiang**, Cheng Lu, Mor Naaman, Daniel Wigdor. The Government's Dividend: Complex Perceptions of Social Media Misinformation in China. *In Proceedings of the 38th Annual SIGCHI Conference on Human Factors in Computing Systems* (***CHI2020***). [PDF]

[1] **Yue Jiang**, Ruofei Du, Christof Lutteroth, Wolfgang Stuerzlinger. ORCLayout: Adaptive GUI Layout with OR-Constraints. *In Proceedings of the 37th Annual SIGCHI Conference on Human Factors in Computing Systems* (***CHI2019***). [PDF]

Workshop Papers, Extended Abstracts, and Doctoral Consortium

[19] Thomas Langerak, Patrick Ebel, Mark Colley, Philipp Wintersberger, Mohamed Chetouani, Jason Wu, Joongi Shin, **Yue Jiang**, Antti Oulasvirta. ICI: An Introduction to Computational Interaction. *In Proceedings of the ACM International Conference on Intelligent User Interfaces* (***ACM IUI 2027 Tutorial***).

[18] Eunhye Kim, Meng Chen, **Yue Jiang**. Task-Aligned Layered Evaluation for Generative Visualizations. *In Proceedings of the IEEE VIS Workshop on Visualization and Generative AI* (***IEEE VIS 2026 Workshop Paper***).

[17] Duotun Wang, Rui Cao, Kunyu Feng, Wanling Li, Qian Zhong, Zhijing Shao, **Yue Jiang**, Zhiming Hu, Mingming Fan, Zeyu Wang. Vision-Language Models with In-Context Learning for In-Camera Portrait Composition Guidance. *In Adjunct Proceedings of the ACM Symposium on User Interface Software and Technology* (***UIST2026 Poster***).

[16] Eunhye Kim, DaEun Choi, Bryan Min, Hyunjung Yi, **Yue Jiang**, Juho Kim. Information Architecture as a Shared Language for Structuring Generative User Interfaces. *In Adjunct Proceedings of the ACM Symposium on User Interface Software and Technology* (***UIST2026 Demo***).

[15] Yao Wang, Junichi Nagasawa, Danqing Shi, Chuhan Jiao, **Yue Jiang** (corresponding author), Andreas Bulling. Is this it? Benchmarking Scanpath Metrics for Information Display. *In Proceedings of Eye Tracking Metrics Workshop (METR)* (***ETRA2026***).

[14] Kewen Peng, Jeffrey Nichols, Christof Lutteroth, Tiffany Knearem, Felix Kretzer, Jeffrey Bigham, Alexander Maedche, **Yue Jiang**. Human-AI-UI Interactions Across Modalities. (***CHI2026 Extended Abstract***).

[13] **Yue Jiang***, Yuwen Lu*, Tiffany Knearem, Clara E Kliman-Silver, Christof Lutteroth, Jeffrey Nichols, Wolfgang Stuerzlinger. Designing and Developing User Interfaces with AI: Advancing Tools, Workflows, and Practices. (***CHI2025 Extended Abstract***).

[12] Kashumi Madampe, John Grundy, Judith Good, Dulaji Hidellaarachchi, JÃącome Cunha, Chris Brown, Peng Kuang, Reham Al Tamime, Ariful Islam Anik, Advait Sarkar, Wei Zhou, Shawal Khalid, Tommaso Turchi, Shavindra Wickramathilaka, **Yue Jiang**. Addressing the agony of recruitment for human-centric computing studies. (***ACM SIGSOFT Software Engineering Notes***).

[11] Tao Long, Sitong Wang, Emilie Fabre, Tony Wang, Anup Sathya, Jason Wu, Savvas Petridis, Dingzeyu Li, Tuhin Chakrabarty, **Yue Jiang**, Jingyi Li, Tiffany Tseng, Ken Nakagaki, Qian Yang, Nikolas Martelaro, Jeffrey V. Nickerson, Lydia B. Chilton. Facilitating Longitudinal Interaction Studies of AI Systems. (***UIST2025 Extended Abstract***).

[10] **Yue Jiang**. Computational Representations for Graphical User Interfaces. (***CHI2024 Doctoral Consortium***).

[9] **Yue Jiang**, Eldon Schoop, Amanda Swearngin, Jeffrey Nichols. ILuvUI: Instruction-tuned LangUage-Vision modeling of UIs from Machine Conversations. (***CHI2024 Workshop Paper***).

[8] Busra Asan, Lena Hegemann, **Yue Jiang (project advisor)**, Antti Oulasvirta. Suggesting Colors for UI Designs with Graph Neural Networks. (***CHI2024 Workshop Paper***).

[7] **Yue Jiang***, Yuwen Lu*, Tiffany Knearem, Clara E Kliman-Silver, Christof Lutteroth, Toby Jia-Jun Li, Jeffrey Nichols, Wolfgang Stuerzlinger. Computational Methodologies for Understanding, Automating, and Evaluating User Interfaces. (***CHI2024 Extended Abstract***).

[6] **Yue Jiang**, Yuwen Lu, Christof Lutteroth, Toby Jia-Jun Li, Jeffrey Nichols, Wolfgang Stuerzlinger. The Future of Computational Approaches for Understanding and Adapting User Interfaces. (***CHI2023 Extended Abstract***).

[5] Yao Wang, Ludwig Sidenmark, Teresa Hirzle, **Yue Jiang**, Andreas Bulling. 8th International Workshop on Pervasive Eye Tracking and Mobile Eye-based Interaction (PETMEI). (***ETRA2023 Extended Abstract***).

[4] Garreth Tigwell, **Yue Jiang**. Disability Semantics for Academic Writing. ***AccessSIGCHI***, 2023.

[3] **Yue Jiang**, Luis A. Leiva, Hamed Rezazadegan Tavakoli, Paul R. B. Houssel, Julia Kylmala, Antti Oulasvirta. UEyes: An Eye-Tracking Dataset across User Interface Types. (***CHI2023 Workshop Paper***).

[2] Lena Hegemann, **Yue Jiang**, Joon-Gi Shin, Yi-Chi Liao, Markku Laine, Antti Oulasvirta. Computational Assistance for User Interface Design: Smarter Generation and Evaluation of Design Ideas. (***CHI2023 Interactivity***).

[1] **Yue Jiang***, Yuwen Lu*, Jeffrey Nichols, Wolfgang Stuerzlinger, Chun Yu, Christof Lutteroth, Yang Li, Ranjitha Kumar, Toby Jia-Jun Li. Computational Approaches for Understanding, Generating, and Adapting User Interfaces. (***CHI2022 Extended Abstract***).

Organized Workshops

[8] Kewen Peng, Beichen Xue, Meng Chen, Felix Kretzer, Jeffrey Nichols, Clara Kliman-Silver, Tiffany Knearem, Alexander Maedche, Christof Lutteroth, Hans Gellersen, **Yue Jiang**. Beyond Gaze: Human and AI Agent Attention in Interfaces. (***CHI2027***). (**under review**)

[7] Kewen Peng, Jeffrey Nichols, Christof Lutteroth, Tiffany Knearem, Felix Kretzer, Jeffrey Bigham, Alexander Maedche, **Yue Jiang**. Human-AI-UI Interactions Across Modalities. (***CHI2026***).

[6] **Yue Jiang***, Yuwen Lu*, Tiffany Knearem, Clara E Kliman-Silver, Christof Lutteroth, Jeffrey Nichols, Wolfgang Stuerzlinger. Designing and Developing User Interfaces with AI: Advancing Tools, Workflows, and Practices. (***CHI2025***).

[5] Tao Long, Sitong Wang, ÃĽmilie Fabre, Tony Wang, Anup Sathya, Jason Wu, Savvas Petridis, Dingzeyu Li, Tuhin Chakrabarty, **Yue Jiang**, Jingyi Li, Tiffany Tseng, Ken Nakagaki, Qian Yang, Nikolas Martelaro, Jeffrey V. Nickerson, Lydia B. Chilton. Facilitating Longitudinal Interaction Studies of AI Systems. (***UIST2025***).

[4] **Yue Jiang***, Yuwen Lu*, Tiffany Knearem, Clara E Kliman-Silver, Christof Lutteroth, Toby Jia-Jun Li, Jeffrey Nichols, Wolfgang Stuerzlinger. Computational Methodologies for Understanding, Automating, and Evaluating User Interfaces. (***CHI2024***).

[3] **Yue Jiang**, Yuwen Lu, Christof Lutteroth, Toby Jia-Jun Li, Jeffrey Nichols, Wolfgang Stuerzlinger. The Future of Computational Approaches for Understanding and Adapting User Interfaces. (***CHI2023***).

[2] Yao Wang, Ludwig Sidenmark, Teresa Hirzle, **Yue Jiang**, Andreas Bulling. 8th International Workshop on Pervasive Eye Tracking and Mobile Eye-based Interaction (PETMEI). (***ETRA2023***).

[1] **Yue Jiang***, Yuwen Lu*, Jeffrey Nichols, Wolfgang Stuerzlinger, Chun Yu, Christof Lutteroth, Yang Li, Ranjitha Kumar, Toby Jia-Jun Li. Computational Approaches for Understanding, Generating, and Adapting User Interfaces. (***CHI2022***).

Academic Service

Conference Organizing Committees

2027 **Doctoral Symposium Chair**, VL/HCC2027

2027 **Workshop and Tutorial Chair**, IUI2027

2026 **Publicity Chair**, VL/HCC2026

2024 **Accessibility Co-Chair**, CHI2024

2023 **Accessibility Co-Chair**, CHI2023

2023 **Co-Organizer & Poster Chair**, HelsinCHI Symposium 2023

2023 **Organizing Committee Member & Keynote Chair**, ELLIS Doctoral Symposium 2023

2022 **Co-Organizer**, ACM SIGCHI Conference on Designing Interactive Systems (DIS2022) Ask-Me-Anything

Program Committees and Editorial Service

2027 **Associate Chair (AC)**, Full Paper, CHI2027

2026 **Associate Chair (AC)**, Full Paper, UIST2026

2026 **Associate Chair (AC)**, Full Paper, CHI2026

2026 **Program Committee Member**, Full Paper, IUI2026

2025 **Program Committee Member**, Full Paper, VL/HCC2025

2025 **Associate Chair (AC)**, Full Paper, CHI2025

2025 **Program Committee Member**, Full Paper, IUI2025

2025 **Associate Chair (AC)**, Poster and Demo, UIST2025

2024 **Committee Member**, ASSETS 2024 Experience Reports

2024 **Associate Chair (AC)**, CHI2024 Late-Breaking Work (SIGCHI Short Paper)

2023 **Program Committee Member**, AI and HCI Workshop, ICML2023

2023 **Workshop Juror**, CHI2023 Workshop

2023 **Associate Chair (AC)**, CHI2023 Late-Breaking Work (SIGCHI Short Paper)

2023 **Program Committee Member**, IUI2023

2022 **Associate Chair (AC)**, CHI2022 Late Breaking Work (SIGCHI Short Paper)

2021 **Program Committee Member (PC)**, ACM IUI2021 Demos and Posters

2021 **Associate Chair (AC)**, CHI2021 Late-Breaking Work (SIGCHI Short Paper)

2020 **Associate Chair (AC)**, CHI2020 Late-Breaking Work (SIGCHI Short Paper)

Conference Session Chairs

2026 **Session Chair**, Designing Interfaces and Graphics with AI, UIST2026

2026 **Session Chair**, Gaze as Input, CHI2026

2025 **Session Chair**, Moving and Looking, CHI2025

2025 **Session Chair**, Pointing and Selection, CHI2025
2024 **Session Chair**, Gaze Interaction in Immersive Environments, CHI2024
2024 **Session Chair**, AI and UI Design, CHI2024
2023 **Session Chair**, Human AI Collaboration, CHI2023
2023 **Session Chair**, Text Input and Textual Communication, CHI2023
2021 **Session Chair**, Interaction and Touch Session, MobileHCI

University and Institutional Service

2026 **Institute Council Member**, Institute for Digital Experience, University of Utah
2026 **Committee Member**, Distinguished Lecture Series Committee, University of Utah
2026 **Graduate Admission Committee Member**, University of Utah
2025 - 2026 **Establishment Committee Member**, Undergraduate AI Major Establishment Committee, University of Utah
2025 **Graduate Admission Committee Member**, University of Utah
2025 **Panelist**, Grad School Open House, University of Utah
2024 **HCI Master Program Admission Committee Member**, Aalto University, Finland
2024 **Ph.D. Admission Committee Member**, Finnish AI Center, Finland
2023 **Ph.D. Admission Committee Member**, Finnish AI Center, Finland
2021 **Ph.D. & Master Admission Committee Member**, University of Maryland, College Park, USA
2020 **Ph.D. & Master Admission Committee Member**, University of Maryland, College Park, USA
2019 **Ph.D. & Master Admission Committee Member**, University of Maryland, College Park, USA
March 2019 **Ph.D Student Panel Leader for Prospective Students**, University of Maryland, USA
2016 **Department of Mathematics Ambassador**, University of Toronto, Canada

Professional Community Leadership and DEI

2025 - Now **Committee Member**, (Re)presented Initiative, Human-Computer Interaction Community
2023 - Now **Member**, ACM AccessSIGCHI
2021 - 2023 **Committee Member**, ACM SIGGRAPH Research Development Committee — DEI and Accessibility
2021 - 2025 **Co-Organizer**, Seminar of HCI for Chinese HCI Researchers around the World
Nov 2019 **Graduate Mentor for Technica Research Bootcamp (Female Student Hackathon)**, USA
Nov 2018 **Graduate Mentor for Technica Research Bootcamp (Female Student Hackathon)**, USA

Academic Reviews and Awards (Reviewed 226 submissions)

2026 **Special Recognitions for Outstanding Reviews for UIST2026**
2026 **2 x Special Recognitions for Outstanding Reviews for CHI2026**
2024 **Special Recognitions for Outstanding Reviews for CHI2024**
2022 **Special Recognitions for Outstanding Reviews for UIST2022**
2021 **Special Recognitions for Outstanding Reviews for CHI2021**
2020 - 2027 SIGCHI Conference on Human Factors in Computing Systems (CHI)
2020 - 2026 ACM User Interface Software and Technology Symposium (UIST)
2026 ACM Symposium on Eye Tracking Research & Applications (ETRA)
2022 - 2026 ACM Transactions on Computer-Human Interaction (TOCHI)
2024 - 2025 Transaction on Visualization and Computer Graphics (TVCG)
2022 - 2025 SIGCHI Workshop paper
2019 - 2024 SIGCHI Late Breaking Work (CHI LBW)
2023 SIGCHI Workshop proposal

2023 ICML Workshop paper
2022 ACM SIGGRAPH
2022 ACM SIGGRAPH Asia
2022 International Symposium on Mixed and Augmented Reality (ISMAR)
2022 ACM Multimedia (MM)
2021 Computers & Graphics (C&C)
2020 Australian Computer-Human Interaction Conference (OzCHI)
2020 ACM International Conference on Interactive Surfaces and Spaces (ISS)
2020 ACM Symposium on Virtual Reality Software and Technology (VRST)
2020 IEEE Virtual Reality Conference (IEEE VR)
2019 IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)
2019 Pacific Graphics (PG)
2019 IFIP Conference on Human-Computer Interaction (INTERACT)
2019 Graphics Interface (GI)

Awards and Honors

2026 **NVIDIA Academic Grant Program Award**
2025 **Aalto Best Dissertation Award**
2023 - 2025 **Meta Research PhD Fellowship**
2022 **Google Europe Students with Disabilities Scholarship**
2024 **Heidelberg Laureate Forum Young Researcher**
2024 **Nokia Scholarship**
2024 **UCSD Data Science Rising Star**
2021 - 2026 **Special Recognitions for Outstanding Reviews x 6 (CHI2021, UIST2022, CHI2024, CHI2026, UIST2026)**
2024 Finnish Center for Artificial Intelligence (FCAI) Travel Grant (22,000 Euros)
2022 - 2025 Finnish Center for Artificial Intelligence (FCAI) PhD Grants
2022 Best Paper Honorable Mention, CHI2022
2022 ACM SIGCHI Gary Marsden Travel Award (In-person)
2021 ACM SIGCHI Gary Marsden Travel Award (Virtual)
2020 Oral Paper Top 5%, CVPR2020
Sept. 2020 Adobe Research Gift Grant for Document Intelligent Research (2020 Fall) ($10,000)
May 2020 Adobe Research Gift Grant for Document Intelligent Research (2020 Summer) ($10,000)
2019 Jacob K. Goldhaber Travel Grant Award
2019 Department of Computer Science Travel Grant Award, University of Maryland
2018 - 2020 Graduate Dean's Scholarship, University of Maryland, College Park ($10,000)
Sept. 2018 Maryland Center For Women In Computing (MCWIC) Grace Hopper Scholarship
2014 - 2018 Dean's List Scholar x 4, University of Toronto
2014 - 2016 New College Council In-Course Scholarship x 2, University of Toronto
2014 - 2015 Award for Outstanding Achievement in Mathematical Expression and Reasoning for Computer Science, University of Toronto
2012 National Olympiad in Biology in Provinces, First Prize
2011 China Adolescents Science and Technology Innovation Contest, First Prize

Teaching Experience

Aug 2026 **Instructor** - **AI for Human-Centered Computing**
- Dec 2026 University of Utah

Jan 2026 **Instructor** - **AI for Human-Centered Computing** **(new course)**
- May 2026 University of Utah
Instructor Evaluation: Undergraduate: 5.8/6; Graduate: 5.68/6

Oct 2025 **Guest Lecturer** - Topic: **Computational Representations for Graphical User Interfaces**
University of Rochester

Oct 2024 **Guest Lecturer** - Topic: **Visual Saliency**
Aalto University

Apr 2024 **Guest Lecturer** - Topic: **Understanding UIs and How People Interact with Them**
University of Rochester

May 2023 **Guest Lecturer** - Topic: **Visual Perception**
Aalto University

Jan 2023 **Research Project Mentor** - **Research Project Course in Human-Computer Interaction**
– May 2023 Aalto University
Instructor: Prof. Antti Oulasvirta

Nov 2022 **Guest Lecturer** - Topic: **Visual Saliency**
Aalto University

Oct 2021 **Co-Instructor** - **Seminar: Computing the User and their interface**
– Feb 2022 Saarland University
Instructors: Prof. Anna Maria Feit, Yue Jiang, Kevin Baum, Markus Langer

Apr 2021 **Co-Supervisor** - **Seminar: Computer Vision and Machine Learning for Computer Graphics**
– Aug 2021 Max Planck Institute for Informatics
Instructors: Prof. Christian Theobalt, Dr. Mohamed Elgharib, Dr. Vladislav Golyanik

Jan 2020 **Teaching Assistant** - CMSC740 **Advanced Computer Graphics**
– May 2020 Department of Computer Science, University of Maryland
Instructor: Prof. Matthias Zwicker

Aug 2019 **Teaching Assistant** - CMSC427 **Computer Graphics**
– Dec 2019 Department of Computer Science, University of Maryland
Instructor: Prof. Matthias Zwicker

Aug 2018 **Teaching Assistant** - CMSC427 **Computer Graphics**
– Dec 2018 Department of Computer Science, University of Maryland
Instructor: Prof. Matthias Zwicker

Jan 2017 **Teaching Assistant** - CSC263 **Data Structures and Analysis**
– Apr 2017 Department of Computer Science, University of Toronto
Instructors: Prof. Sam Toueg and Prof. Francois Pitt

Sept 2016 **Teaching Assistant** - CSC263 **Data Structures and Analysis**
– Dec 2016 Department of Computer Science, University of Toronto
Instructor: Prof. David Liu

Jan 2016 **Teaching Assistant** - CSC165 **Mathematical Expression and Reasoning for Computer Science**
– Apr 2016 Department of Computer Science, University of Toronto
Instructors: Dr. Ilir Dema and Dr. Abdallah Farraj

Sept 2015 **Peer Tutor** - MAT223/224 **Linear Algebra I & II**

– Apr 2016 Department of Mathematics, University of Toronto

Sept 2015 **Peer Tutor** - MAT137 **Calculus**
– Apr 2016 Department of Mathematics, University of Toronto

2014 **Teaching Volunteer** - High School **Mathematics** and **Physics**
Qinyuan, Shanxi Province, China

Mentoring Experience

Ph.D. Students:

2026 - Now Kewen Astrid Peng

2026 - Now Beichen Xue (soon to be a PhD student)

Undergraduate and Master Students:

2026 Zicong Geng (Undergraduate Research Assistant)

2026 Chenming Ge (Undergraduate Research Assistant)

2026 Chengyang Shi (Undergraduate Research Assistant)

2026 Jewan Chae (Undergraduate student, University of Utah)

2026 Parth Desai (Master's student, University of Utah)

2026 Kaden Salem (BS/MS student, University of Utah)
Next Stop: Software Engineer @ Goldman Sachs

2026 Yura Hwang (Graduate Research Assistant, University of Utah)
Next Stop: Ph.D. Student @ University of Utah

2026 Bingkun Han (Undergraduate student, University of Utah)
Next Stop: Northeastern University

2024 - 2026 Aryan Garg (Undergraduate student, visiting from IIT)
Topic: Visual Saliency Informed Controlled Synthesis of User Interfaces
Next Stop: Ph.D., University of Wisconsin-Madison

2024 Amanda Li (Master's student, CMU)
Topic: Image Editing
Next Stop: Ph.D., Stanford University

2024 Elise Chenxinran Shen (Master's student, University of British Columbia)
Topic: Simulating Human Behaviors on Social Media
Next Stop: Ph.D., University of Toronto

2023 Ananya Dutta (Undergraduate student, Manipal University Jaipur)
Topic: Scanpath-based UI Optimization
Next Stop: Google

2023 Changkong Zhou (Master's student, Aalto University)
Topic: UI Autocompletion
Next Stop: Alibaba

2023 Busra Asan (Master's student, Istanbul Technical University)
Topic: Color Optimization for UIs
Next Stop: ML Master's Student, University of Tubingen

2023 Henrik Kauppi (Master's student, Aalto University)
Topic: Scanpath Dataset for UI Transitions

2023 Lotta Merisaari (Master's student, Aalto University)
Topic: Scanpath Dataset for UI Transitions

2023 Yao Zhang (Master's student, Aalto University)

Topic: UI Embedding

2023 Elham Sadri (Master's student, Simon Fraser University)

Topic: Design and Evaluation of a New Editor for Responsive Graphical User Interfaces

2021 Kartik Teotia (Master's student, Max Planck Institute)

Topic: NeRF and Signed Distance Field (SDF)

Next Stop: Ph.D., Max Planck Institute for Informatics

2020 Duotun Wang (Master's student, University of Maryland)

Topic: Signed Distance Field-Based Differentiable Sketching

Next Stop: Ph.D., Hong Kong University of Science and Technology, Guangzhou Campus

2018, 2019 Graduate Mentor for Technica (All-Women Hackathon)

2018 - 2020 Graduate Mentor at Maryland Center For Women In Computing (MCWIC) Peer mentoring

Invited Talks

June 2026 **The Synthetic User: Simulating Viewing Behaviors through Multimodal AI**, *Keynote Talk* at the Generative AI meets Eye Tracking (GenEAI) Workshop, ETRA2026

Apr 2026 **How can Agents Help us?**, *Keynote Talk* at the Human-AI-UI Interactions Across Modalities Workshop, CHI2026

Nov 2025 **Computational Understanding of User Interfaces**, "Research Challenges in Computing" Seminar, University of Utah, USA

Jan - Apr, 2025 **Computational Representations for User Interfaces**, University of Cambridge, Chinese University of Hong Kong (Shenzhen Campus), Apple AI/ML Lab, Hong Kong University of Science and Technology (Guangzhou Campus), University of Maryland Baltimore County, Simon Fraser University, University of Alberta, William & Mary, Max Planck Institute, York University, University of Sydney, Swiss Federal Technology Institute of Lausanne (EPFL), University of Washington, University of Virginia, University of Tennessee, Lehigh University, University of Utah, University of Victoria

Feb, 2025 **Computational Representations for User Interfaces**, University of Chicago (Hosted by Pedro Lopez)

Oct 2024 **Toward Computational User Interfaces**, University of Waterloo, Canada (Hosted by Dan Vogel)

Oct 2024 **Toward Computational User Interfaces**, University of Toronto, Canada (Hosted by Fanny Chevalier)

Oct 2024 **Effective Computational Representation for AI-Assisted UI Design**, Columbia University, USA (Hosted by Brian A. Smith)

Aug 2024 **Revolutionizing UI Design: Computational Representations and AI Assistance**, University of Illinois Urbana-Champaign, USA (Hosted by Yun Huang)

May 2024 **Computational UI Representation**, Stanford University, USA (Hosted by Yujie Tao)

Apr 2024 **Understanding UIs and How People Interact with Them**, Massachusetts Institute of Technology, USA (Hosted by Stefanie Mueller)

Apr 2024 **Understanding UIs and How People Interact with Them**, University of Rochester, USA (Hosted by Yukang Yan)

Apr 2024 **Understanding UIs and How People Interact with Them**, Northeastern University, USA (Hosted by Dakuo Wang)

June 28, 2023 **Computational Approaches for UI Understanding**, Google, USA (Hosted by Tiffany Knearem)

Apr 13, 2023 **UEyes: Understanding Visual Saliency across User Interface Types**, HelsinCHI Symposium, Finland (Hosted by Antti Oulasvirta)

Apr 13, 2023 **Demo: Computational Assistance for User Interface Design: Smarter Generation and Evaluation of Design Ideas**, HelsinCHI Symposium, Finland (Hosted by Antti Oulasvirta)

Apr 12, 2023 **Computational Approaches for User Interfaces**, City University of Hong Kong, China (Hosted by Zhicong Lu)

Apr 12, 2023 **UEyes: Understanding Visual Saliency across User Interface Types**, CHI Lux Seminar, Luxembourg (Hosted by Luis Leiva)

Aug 24, 2022 **Computational Design – Optimization for Adaptive User Interfaces**, DELTA Summer Workshop 2022, Finland (Hosted by Jari Nurmi)

Apr 8, 2022 **Adaptive User Interfaces**, Hasso Plattner Institute, Germany (Hosted by Ran Zhang & Patrick Paudisch)

Mar 17, 2022 **Adaptive User Interfaces**, University of Luxembourg, Luxembourg (Hosted by Luis Leiva)

Dec 7, 2021 **Adaptive User Interfaces and 3D Reconstruction**, ETH Zurich, Switzerland (Hosted by Christian Holz & Otmar Hilliges)

Sept 27, 2021 **Adaptive User Interface**, Seminar of HCI for Chinese HCI Researchers

Dec 2, 2020 **Adaptive GUI Layout**, DGP HCI Session, University of Toronto, Canada (Hosted by DGP Lab)

Sept 4, 2020 **Responsive Document Using OR-Constraint Optimization**, Document Intelligence Lab Talk, Adobe Research, USA (Hosted by Vlad Morariu)

Aug 18, 2020 **High-Speed, High-Accuracy, Low-Latency Touch Tracking**, Apple Annual Showcase for FIGLab, Apple & CMU, USA (Hosted by Chris Harrison)

Aug 18, 2020 **Doppler Radar for Activity Recognition**, Apple Annual Showcase for FIGLab, Apple & CMU, USA (Hosted by Chris Harrison)

July 31, 2020 **High-Speed, High-Accuracy, Low-Latency Touch Tracking**, FIGLab Research Review Presentation, Apple & CMU, USA (Hosted by Chris Harrison)

July 31, 2020 **Doppler Radar for Activity Recognition**, FIGLab Research Review Presentation, Apple & CMU, USA (Hosted by Chris Harrison)

July 14, 2020 **Document Layout with OR-Constraints**, Graphics Intelligence Lab Talk, Adobe Research, USA (Hosted by Paul Asente)

June 27, 2020 **SDFDiff: Differentiable Rendering of Signed Distance Fields for 3D Shape Optimization**, DeeCamp AI Training Camp, AI Institute of Sinovation Ventures, China (Hosted by Ran Zhang (IST Austria))

May 29, 2020 **ORCSolver: An Efficient Solver for Adaptive GUI Layout with OR-Constraints**, German CHI Week, German HCI, Germany (Hosted by Teresa Hirzle & Christina Schnegass)

May 20, 2020 **ORCSolver: An Efficient Solver for Adaptive GUI Layout with OR-Constraints**, BathCHI 2020 Seminar, University of Bath, UK (Hosted by Christof Lutteroth)

Dec 3, 2019 **ORC Layout: Adaptive GUI Layout with OR-Constraints**, Document Intelligence Lab Talk, Adobe Research, USA (Hosted by Vlad Morariu & Tong Sun)

July 11, 2019 **ORC Layout: Adaptive GUI Layout with OR-Constraints**, Visual Computing Summer School 2019, Shenzhen University, China (Hosted by Hui Huang)

May 15, 2019 **Signed Distance Function Based Differentiable Rendering**, Capital Graphics 2019, George Mason University, Arlington, VA, USA (Hosted by Yotam Gingold)

May 2, 2019 **ORC Layout: Adaptive GUI Layout with OR-Constraints**, UMD HCI Lab BBL Talk, University of Maryland, USA

April 4, 2019 **ORC Layout: Adaptive GUI Layout with OR-Constraints**, Human-Computer Interaction Lab 36th Annual Symposium, USA (Hosted by Niklas Elmqvist)

April 1, 2019 **ORC Layout: Adaptive GUI Layout with OR-Constraints**, UMD HCI Lab SIGCHI Paper Talk Session, University of Maryland, USA (Hosted by Hernisa Kacorri)

Jun 21, 2018 **Neural Programmer Interpreter**, Knowledge Representation and Reasoning Talk Session, University of Toronto, Canada (Hosted by Sheila McIlraith)

Nov 8, 2016 **Vector Addition Systems Reachability Problem**, Mathematical Linguistics Talk Session, University of Toronto, Canada (Hosted by Gerald Penn)

Oct 25, 2016 **Supertagging: A Non-Statistical Parsing-Based Approach**, Mathematical Linguistics Talk Session, University of Toronto, Canada (Hosted by Gerald Penn)

Oct 3, 2016 **Reachability Problems for Vector Addition Systems**, University of Toronto Undergraduate Computer Theory Talk Session, University of Toronto, Canada

July 20, 2016 **Reachability Problems and Vector Addition Tree Automata**, Undergraduate Summer Research Program (UGSRP) Talk, University of Toronto, Canada

Press Coverage

May 11, 2021 **CMU researchers show potential of privacy-preserving activity tracking using radar** – TechCrunch

May 8, 2019 **Eases the pain of multiple UI designs** – IT Works Solution

May 7, 2019 **New open source software eases the pain of multiple UI designs** – Phys.org

May 7, 2019 **New software eases the pain of multiple UI designs** – University of Bath, UK

Voluntary Experience

Oct 2025 Student Volunteer for the ACM International Conference on Ubiquitous Computing **(Ubicomp2025)**, Espoo, Finland.

May 2022 Student Volunteer for the SIGCHI Conference on Human Factors in Computing Systems **(CHI2022)**, New Orleans, USA.

Oct 2021 Student Volunteer for the ACM Symposium on User Interface Software and Technology **(UIST2021)** (Virtual).

Sept 2021 Student Volunteer for the ACM International Conference on Mobile Human-Computer Interaction **(MobileHCI2021)**, Toulouse, France (Virtual).

Nov 2020 Student Volunteer for the ACM Interactive Surfaces and Spaces **(ISS2020)**, Lisbon, Portugal (Virtual).

April 2019 Student Volunteer for **Human-Computer Interaction Lab 36th Annual Symposium**, Maryland, USA

May 2016 Volunteer for **Doors Open Toronto 2016**, City Cultural Events, Toronto, Canada